

Grading — Specs Model

INFO 521 · Machine Learning Foundations · specs grading reference

How to read this

This is the reference for how project work is graded: specifications, not points. It governs the mechanics; the category weights live in the syllabus. Milestone notebooks carry their own Satisfactory criteria, and the checkpoint quizzes run on D2L.

Course-schedule placement. Reference document. Applies to every project milestone and checkpoint quiz across both projects; no single week.

Both projects use specifications grading. There are no points on milestones; each is judged against published criteria as **Satisfactory** or **Not-Yet**.

Milestones

- Every milestone lists its Satisfactory criteria at the top of the notebook. All criteria must be met — it is all-or-nothing, not partial credit.
- A **Not-Yet** returns with targeted feedback and **one revise cycle** back to Satisfactory. (Set a revise window that fits your cadence, e.g. one week.)
- A project is complete when all its milestones are Satisfactory **and** all its checkpoints are passed.

Checkpoints

- Short open-book **D2L quizzes** (about 20 to 30 minutes), one per module, on that module's derivations. Mostly auto-graded, plus two or three short written responses.
- **Satisfactory** requires the auto-graded pass bar (80%) and Satisfactory written responses; written responses carry **one revision** cycle, matching the milestone policy.
- Each quiz is due the week its module is taught and **gates the milestone due the following week**, so the gate resolves before the gated work is due.
- The coding verification that used to sit beside these derivations lives in Homework Units 1, 3, and 4 and in milestones 2.1, 2.3, and 2.4 (see `info521-homeworks-2026/CHECKPOINT_HOMEWORK_MERGE.md`). Nothing is assessed twice.
- A milestone cannot be marked Satisfactory until its gating checkpoint quiz is passed.

One project, two parts

The course has **one project**, in two parts on one dataset. Part 1, *One Problem, Three Lenses* (milestones 1.1 to 1.4), is due **Week 5, the midpoint**. Part 2, *From Inference to Discovery* (milestones 2.1 to 2.5), is due **Week 7.5**. Gated as below.

All five checkpoint quizzes are required as the Checkpoints category, and each gates the milestone it is named for.

Weighting

Part 2 is **not** weighted more heavily than Part 1: specs grading already enforces that Part 2 mastery presupposes Part 1's foundations, so there is no need to also price that into a weight.

Revise-and-resubmit is the safety net; it replaces the former "final replaces lowest midterm" rule cleanly.

The exact contribution of each project to the final grade depends on how these fold into the broader course scheme (homework units, peer-engagement loops). Set that in the syllabus; this file governs the *mechanics*, not the percentages.

Bundling (suggested)

- **Part 1 bundle:** 1.1, 1.2, 1.3, 1.4 all Satisfactory + Checkpoint quizzes 1.1 to 1.3 passed. Due at the midpoint (Week 5).
- **Part 2 bundle:** two thematic gates + the synthesis, all Satisfactory, plus Checkpoint quizzes 2.1 and 2.3 passed:
 - **Gate A — Supervised:** 2.1 (approximate inference) + 2.2 (classification) Satisfactory, **and Checkpoint 2.1** (Newton–Raphson update) passed.
 - **Gate B — Unsupervised:** 2.3 (clustering) + 2.4 (dimensionality reduction) Satisfactory, **and Checkpoint 2.3** (k-means objective / PCA) passed.
 - **Synthesis:** 2.5 Satisfactory, with its structured peer review completed.
- **The project** is Satisfactory when both part bundles are.

Academic integrity note

Homework is open-book and AI-permitted in this course, so it certifies "can produce with help." The checkpoint quizzes are delivered online and are also open-book, so they do **not** certify unaided derivation, and this document no longer claims they do. What they certify is narrower and still worth having: that a student can produce the derivation, state what it depends on, and connect it to their own project results before the milestone that uses it comes due.

The load-bearing parts are the written responses, because they ask about the student's own fits and results. That tie to their own project is what makes the answer theirs rather than generic. Keep it intact if you adjust the model.